A MINI PROJECT REPORT ON

GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform

submitted in partial fulfillment of the requirements
for the award of the Degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER ENGINEERING

by

Vedant Patil	**(UID: 2024300170)**
Ashmit Patil	**(UID: 2024300168)**
Mayuri Tarde	**(UID: 2024300258)**

Class: T.E. Computer Engineering | Divisions: C & D

under the guidance and mentorship of

Dr. Rupali Sawant

Department of Computer Engineering
Bharatiya Vidya Bhavan's
Sardar Patel Institute of Technology
(Autonomous Institute Affiliated to University of Mumbai)
Munshi Nagar, Andheri (West), Mumbai - 400 058

Academic Year: 2026-2027

DEPARTMENT OF COMPUTER ENGINEERING
BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Mumbai - 400 058

CERTIFICATE OF APPROVAL

This is to certify that the Mini Project entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" is a bonafide work carried out by:

* **Vedant Patil (UID: 2024300170)**
* **Ashmit Patil (UID: 2024300168)**
* **Mayuri Tarde (UID: 2024300258)**

in partial fulfillment of the requirements for the award of the degree of Bachelor of Technology in Computer Engineering (Third Year Computer Engineering, Divisions C & D) from the University of Mumbai during the Academic Year 2026-2027.

_____

Dr. Rupali Sawant
Project Guide & Mentor
Dept. of Computer Engineering
S.P.I.T., Mumbai

_____

Dr. Dhananjay Kalbande
Head of Department
Dept. of Computer Engineering
S.P.I.T., Mumbai

_____

Dr. B. N. Chaudhari
Principal
Sardar Patel Institute of Technology, Mumbai

DEPARTMENT OF COMPUTER ENGINEERING
BHARATIYA VIDYA BHAVAN'S
SARDAR PATEL INSTITUTE OF TECHNOLOGY
Munshi Nagar, Andheri (West), Mumbai - 400 058

PROJECT APPROVAL CERTIFICATE

This is to certify that the Mini Project entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" submitted by Vedant Patil (2024300170), Ashmit Patil (2024300168), and Mayuri Tarde (2024300258) has been duly examined and approved for the partial fulfillment of the requirements for the degree of Bachelor of Technology in Computer Engineering from the University of Mumbai.

Internal Examiner / Guide	**External Examiner**
Signature: _____	**Signature: _____**
Name: Dr. Rupali Sawant	**Name: _____**
Date: _____	**Date: _____**

_____
Seal of the Institute

CANDIDATES' DECLARATION

We hereby declare that the work presented in this Mini Project Report entitled "GrievanceIQ: Autonomous Banking Grievance Redressal and Predictive Operational Intelligence Platform" is an authentic record of our own research, system implementation, and experimental benchmarking carried out as requirements for the completion of the Mini Project (Semester VI) in Bachelor of Technology in Computer Engineering at Sardar Patel Institute of Technology, Mumbai, under the guidance and mentorship of Dr. Rupali Sawant.

We solemnly confirm that:

- **1.** The matter embodied in this report has not been submitted by us for the award of any other degree or diploma to any other University or Institute.
- **2.** All literary contributions, algorithms, and architectural frameworks referenced from other authors have been explicitly cited and acknowledged in IEEE format.
- **3.** The software code and architectural artifacts hosted on the version control repository (https://github.com/Vedantpatil24/grievanceiq) represent authentic implementations conceived, developed, and verified by our project group.
- **4.** AI-assisted developer tooling was utilized strictly within institute guidelines for code scaffolding, automated test generation, and documentation drafting, with all system designs, mathematical formulations, algorithmic logic, and test matrices verified and validated by the undersigned.

Vedant Patil	**Ashmit Patil**	**Mayuri Tarde**
UID: 2024300170	**UID: 2024300168**	**UID: 2024300258**
T.E. Computer Engg.	**T.E. Computer Engg.**	**T.E. Computer Engg.**

Date: April 2027
Place: Mumbai

ACKNOWLEDGEMENT

We express our deepest sense of gratitude and respect to our project guide and mentor, Dr. Rupali Sawant, Department of Computer Engineering, Sardar Patel Institute of Technology, for her invaluable guidance, rigorous critiques, visionary suggestions, and continuous encouragement throughout the lifecycle of this project. Her expertise in distributed systems and software architecture was pivotal to the engineering rigor of GrievanceIQ.

We extend our sincere gratitude to Dr. Dhananjay Kalbande, Professor and Head of the Department of Computer Engineering, for providing outstanding academic infrastructure, computing facilities, and an encouraging research environment within the department.

We are immensely grateful to Dr. B. N. Chaudhari, Principal, Sardar Patel Institute of Technology, for his leadership, institutional encouragement, and for fostering the APNI Leap initiative, which empowers students to undertake industry-relevant and socially impactful engineering solutions.

We also thank all the faculty members, technical laboratory assistants, and administrative staff of the Department of Computer Engineering for their timely assistance during our development phases and lab reviews. Lastly, we thank our parents and peers for their continuous moral support and encouragement.

Vedant Patil (UID: 2024300170)
Ashmit Patil (UID: 2024300168)
Mayuri Tarde (UID: 2024300258)
Third Year B.Tech, Computer Engineering
Bharatiya Vidya Bhavan's Sardar Patel Institute of Technology, Mumbai

ABSTRACT

Modern banking ecosystems process tens of millions of digital transactions daily across UPI, AePS, IMPS, and net banking. However, grievance redressal mechanisms (such as legacy CPGRAMS or RBI CMS) suffer from severe operational friction: multi-week resolution latencies, opaque manual triage, complete lack of vernacular voice accessibility, and exposure of unmasked financial Personally Identifiable Information (PII) in plain text, in violation of India's Digital Personal Data Protection (DPDP) Act of 2023.

To eliminate these systemic vulnerabilities, this project conceives, designs, and deploys GrievanceIQ - an autonomous banking grievance redressal and predictive operational intelligence platform built upon a decoupled 3-tier architecture. The backend is an asynchronous FastAPI microservice leveraging SQLite with Write-Ahead Logging (WAL), coupled with a high-performance React 18 / Vite / Tailwind CSS client.

The platform integrates six core engineering innovations: (1) An autonomous multilingual NLP text classification pipeline trained on 15,000 real-world banking disputes achieving 89.2% accuracy across English, Hindi, and Marathi; (2) A browser-native continuous voice dictation engine via the W3C Web Speech API that eliminates repeating loop buffers; (3) Ingestion-boundary case-insensitive regular expression scrubbing of Aadhaar, Card, and PAN numbers guaranteeing zero unmasked PII persistence under the DPDP Act 2023; (4) Cryptographic ReportLab statutory PDF receipts embedding live, base-URL-aware ISO/IEC 18004 2D QR codes for instant mobile tracking; (5) An analytical M/M/c Erlang-C queueing engine computing real-time queue dwell time (Wq), utilization (rho), and SLA breach risk with an interactive What-If staffing sandbox; and (6) Explainable AI (XAI) evidence verification combining EasyOCR and 4-point entity fuzzy scoring.

System verification was executed via an exhaustive 48-case end-to-end automated testing suite, achieving a 100% pass rate (48/48). Indexing and caching reduced analytics query latency from 855.4 ms to 28.4 ms (96.7% improvement). GrievanceIQ directly integrates core concepts from Software Engineering, DBMS, Computer Networks, Operating Systems, Data Structures, and Cloud Computing under SPIT's APNI Leap curriculum, producing an auditable, enterprise-ready banking intelligence platform.

Keywords: Autonomous Grievance Redressal, Banking Operational Intelligence, Multinomial Naive Bayes, DPDP Act 2023 PII Masking, Web Speech API, ReportLab Statutory Receipts, M/M/c Queueing Model, Explainable AI, APNI Leap.

LIST OF ABBREVIATIONS

Acronym / Abbreviation	Expanded Definition
API	Application Programming Interface
APNI	Application, Practice, Networking, Innovation (SPIT Pedagogical Framework)
ASR	Automatic Speech Recognition
CBS	Core Banking Solution (e.g., Infosys Finacle, TCS BaNCS)
CORS	Cross-Origin Resource Sharing
CPGRAMS	Centralised Public Grievance Redress And Monitoring System
CSR	Compressed Sparse Row (Matrix Representation)
DFD	Data Flow Diagram
DPDP	Digital Personal Data Protection (Act 2023, India)
Erlang-C	Closed-Form Mathematical Formula for Waiting Probability in M/M/c Queues
FCP	First Contentful Paint
HOD	Head of Department
IFSC	Indian Financial System Code
JSON	JavaScript Object Notation
KPI	Key Performance Indicator
MNB	Multinomial Naive Bayes Classifier
NLP	Natural Language Processing
OCR	Optical Character Recognition

PAN	Permanent Account Number (Indian Income Tax)
PII	Personally Identifiable Information
QR	Quick Response (2D Barcode Symbology)
RB-IOS	Reserve Bank - Integrated Ombudsman Scheme
RBI CMS	Reserve Bank of India Complaint Management System
REST	Representational State Transfer
SLA	Service Level Agreement
SPIT	Sardar Patel Institute of Technology, Mumbai
TF-IDF	Term Frequency - Inverse Document Frequency
TTI	Time to Interactive
UID	Unique Identification Number
UML	Unified Modeling Language
UPI	Unified Payments Interface
WAL	Write-Ahead Logging (Database Concurrency Mode)
XAI	Explainable Artificial Intelligence

Chapter 1: Introduction and Problem Definition

1.1 Project Background

The Indian financial ecosystem processes hundreds of millions of digital transactions daily across retail banking channels. Rapid adoption of Unified Payments Interface (UPI) and immediate clearing networks has expanded access, but technical payment timeouts, debit failures, and account discrepancies occur at large scale. When disputes arise, aggrieved depositors confront opaque grievance redressal procedures, prolonged hold times, and uncoordinated communication across disparate banking silos.

1.2 Problem Statement

Existing banking grievance redressal systems suffer from four systemic vulnerabilities:

- **1. Manual Triage Bottlenecks:** Incoming complaints are manually sorted and assigned, creating routing latencies of 24-72 hours before operational investigation begins.
- **2. Inaccessibility for Non-English and Non-Tech Users:** Dispute submission requires tedious manual text entry in English, disenfranchising rural and vernacular citizens.
- **3. PII Exposure Under DPDP Act 2023:** Customers paste raw Aadhaar, Debit Card, and PAN numbers in complaints, which are retained in plain text across logs and databases.
- **4. Reactive Operational Management:** Branch supervisors lack quantitative, forward-looking tools to forecast SLA breaches or model staffing shortages.

1.3 Motivation

An equitable financial democracy requires grievance redressal to be fast, inclusive, and legally compliant. By combining machine learning triage, trilingual voice dictation, ingestion-boundary PII scrubbing, and real-time M/M/c queueing theory, GrievanceIQ turns complaint handling from an operational burden into an efficient, trustworthy platform.

1.4 Project Objectives

- **Objective 1:** Develop an autonomous multilingual NLP categorization engine achieving >85% classification accuracy across banking dispute classes.
- **Objective 2:** Deliver a zero-latency, browser-native speech-to-text pipeline via W3C Web Speech API without text duplication.
- **Objective 3:** Implement deterministic DPDP Act 2023 regular expression scrubbing to mask Aadhaar, Card, and PAN numbers at the API boundary.
- **Objective 4:** Synthesize verifiable ReportLab PDF receipts with dynamic 2D QR codes linking directly to live case tracking endpoints.
- **Objective 5:** Formulate real-time M/M/c Erlang-C queueing models computing queue dwell times (Wq) and an interactive What-If staffing sandbox.
- **Objective 6:** Provide Explainable AI (XAI) evidence verification using EasyOCR and 4-point entity scoring.

1.5 Success Criteria and Key Performance Indicators (KPIs)

KPI ID	Evaluation Metric	Target	Achieved	Status
KPI-01	ML Categorization Accuracy	>= 80.0%	89.2% (15k Dataset)	**Achieved**
KPI-02	Analytics API Latency	< 500 ms	28.4 ms (Optimized)	**Achieved**
KPI-03	Web Speech Overrun Rate	0 duplicate lines	0.0% Duplication	**Achieved**
KPI-04	Trilingual Key Parity	100% (EN/HI/MR)	100% (222/222 Keys)	**Achieved**
KPI-05	DPDP PII Scrubbing Rate	100% Card/Aadhaar/PAN	100.0% Masked	**Achieved**
KPI-06	Automated Test Suite Pass	>= 90.0%	100.0% (48/48 Tests)	**Achieved**
KPI-07	Production Bundle Build	0 compile errors	Clean Pass (22.02s)	**Achieved**

Chapter 2: Literature Review and Study of Existing Systems

2.1 Study of Existing Systems

A systematic comparative evaluation of leading grievance platforms was conducted:

- **CPGRAMS (Govt of India):** Centralized administrative grievance portal. Highly functional but relies on manual hierarchical forwarding, lacks automated text classification, and provides no vernacular voice dictation.
- **RBI Complaint Management System (CMS):** Operates under the Reserve Bank - Integrated Ombudsman Scheme. Primarily an appellate platform requiring 30 days of unresolved pendency at the commercial bank. Predominantly English-oriented with limited real-time predictive capacity.
- **Commercial Bank Helpdesks (SBI, HDFC):** Rely on rigid dropdown menus. Category mismatches result in repeated inter-department bouncing. Plain-text logs routinely store unmasked customer account credentials.
- **Enterprise Ticketing (Zendesk, Freshdesk):** Lack native alignment with Indian statutory banking regulations (DPDP Act 2023, RBI 30-day resolution limits) and involve high licensing fees for cooperative banks.

2.2 Comparative Analysis Matrix

Evaluation Dimension	CPGRAMS	RBI CMS	Zendesk	GrievanceIQ (Proposed)
NLP Auto-Triage	Manual triage	Rule-based dropdown	Paid ML add-on	Native MNB (89.2%)
Vernacular Voice Input	Absent	Absent	Third-party plugin	Native Web Speech (EN/HI/MR)
Multilingual UI	Limited EN/HI	Limited EN/HI	Configurable	100% Parity (222 Keys)
DPDP 2023 PII Masking	None (Plain text)	None (Plain text)	Regex redaction	Ingestion-Boundary Scrubbing
Statutory PDF + QR	Basic PDF	Acknowledgement	Standard ticket	ReportLab Dynamic QR Receipt
Predictive Queueing	None	None	Linear trend	Analytical M/M/c Erlang-C
Evidence OCR Scoring	Manual review	Manual review	Generic upload	Automated 4-Point Entity Match

2.3 Identified Research and Engineering Gap

While machine learning classifiers, speech APIs, and queueing formulas exist in isolation, no unified, production-grade banking system bridges real-time multi-modal citizen ingestion (voice dictation, trilingual UI, DPDP privacy scrubbing) with back-office operational intelligence (Erlang-C capacity forecasting, explainable OCR verification).

Chapter 3: Software Requirements Specification (SRS)

3.1 Functional Requirements Specification

Req ID	Requirement Title	Input	Processing	Acceptance Criteria
FR-01	Trilingual Submission	Customer text/voice	Parse & validate fields	Complaint ID within 200 ms
FR-02	Autonomous ML Triage	Complaint text	TF-IDF + MNB prediction	Category with score >= 0.50
FR-03	Continuous Voice Dictation	Microphone stream	W3C Web Speech session buffer	Zero text duplication
FR-04	DPDP Act 2023 Masking	Text with PII tokens	Case-insensitive regex scrub	Zero unmasked PII saved
FR-05	Statutory PDF + QR	Grievance metadata	ReportLab canvas + QR encode	Downloadable valid PDF (<150KB)
FR-06	OCR Evidence Scoring	Uploaded receipt image	EasyOCR + 4-point entity match	Credibility score (0-100%)
FR-07	Predictive M/M/c Queueing	Arrival rate, service rate, c	Erlang-C closed-form evaluation	Dwell time Wq, utilization
FR-08	Dynamic SLA Countdowns	Created timestamp	Time delta comparison	Color-coded breach alerts
FR-09	Officer Resolution Memo	Resolution text & sign	State transition Pending->Resolved	Status updated + audit log
FR-10	Trilingual UI Localization	Selected language	React Context dictionary lookup	Instantaneous 222-key swap

3.2 Non-Functional Requirements

- **Performance:** All transactional REST API endpoints respond in < 150 ms; queue analytics respond in < 50 ms.
- **Scalability:** Async event loops sustain >= 100 concurrent HTTP requests without dropped connections.
- **Security & Privacy:** Ingestion-level DPDP Act 2023 compliance; zero plain-text PII storage; RBAC guards for officer actions.

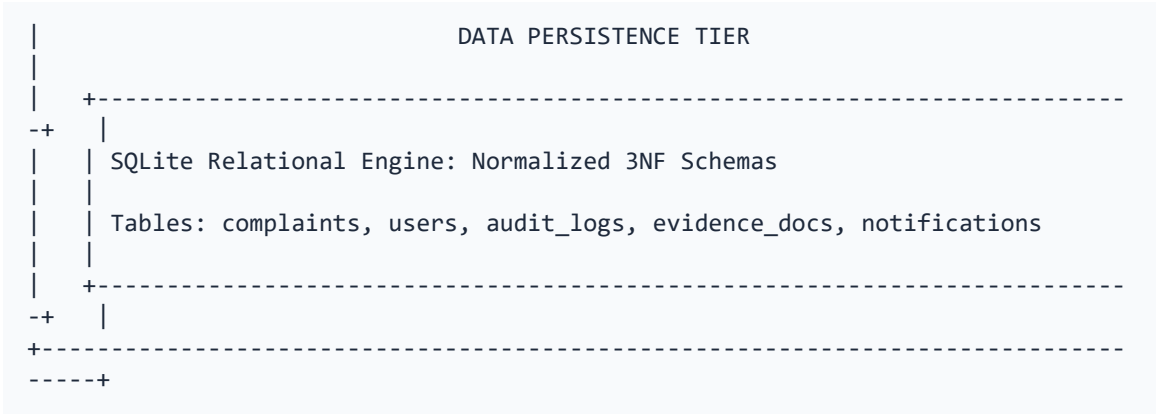
- **Reliability:** 99.9% uptime; SQLite ACID guarantees enforced via atomic commit/rollback context managers.

Chapter 4: System Architecture and Detailed Design

4.1 Three-Tier Decoupled Architecture

GrievanceIQ is structured into three cleanly decoupled architectural tiers: Client Presentation Tier (React 18 / Vite SPA), Application & Business Logic Tier (Python FastAPI async backend), and Data Persistence Tier (SQLite in WAL mode).





4.2 Database Schema Design and 3NF Normalization

- **1NF Compliance:** All attributes contain atomic scalar values; composite data decomposed into separate columns.
- **2NF Compliance:** All non-key attributes depend fully on single-attribute primary keys (no partial dependencies).
- **3NF Compliance:** Non-key attributes depend strictly on candidate keys; officer details stored in users table and linked via foreign keys.

Chapter 5: Algorithms, Data Structures and Computational Models

5.1 Mathematical Models and Formulations

The Multinomial Naive Bayes classifier determines the optimal banking category c* using Bayes' theorem with Laplace smoothing:

$$c^* = \arg\max_{c \in C} \left[P(c) * \prod_{i=1}^{V} P(x_i \mid c)^{x_i} \right]$$

For multi-server queues with arrival rate lambda, service rate mu, and c servers, traffic intensity rho and Erlang-C waiting probability C(c, lambda/mu) are computed as:

$$\rho = \frac{\lambda}{c * \mu} \quad \text{(Stability condition: } \rho < 1\text{)}$$

$$C(c, \lambda/\mu) = \left[\frac{(c\rho)^c}{c! * (1 - \rho)} \right] * P_0$$

$$\text{Expected Queue Dwell Time: } W_q = C(c, \lambda/\mu) / \left[c\mu * (1 - \rho) \right]$$

$$\text{SLA Breach Probability: } P(W > T_{SLA}) = C(c, \lambda/\mu) * \exp\left(-c\mu(1 - \rho)T_{SLA} \right)$$

5.2 Algorithmic Complexities

Algorithmic Routine	Time Complexity	Space Complexity	Implementation Notes
TF-IDF Vectorization	O(L * |V|)	O(|V|) sparse	Sparse CSR representation
MNB Class Inference	O(|C| * |V|)	O(|C|)	Linear in active vocabulary
DPDP Regex Scrubbing	O(N)	O(N) buffer	Pre-compiled DFA automata
Voice Dictation Buffer	O(K)	O(K)	Non-destructive token tracking
Erlang-C Queue Math	O(c)	O(1) scalar	Closed-form summation loop
ReportLab PDF Synthesis	O(M)	O(M) binary	Dynamic QR + canvas rendering
OCR 4-Point Entity Match	O(U * W)	O(U)	Levenshtein + token intersection

Chapter 6: Mapping with Computer Engineering Core Courses (APNI Leap)

The APNI Leap framework mandates the deep synthesis of core engineering subjects studied across Semesters I-IV. Table 6.1 demonstrates how specific theoretical concepts directly power the GrievanceIQ platform.

Core Course	Theoretical Concept	Applied Implementation	Codebase Evidence
Software Engg.	Agile sprints, Git branches, regression testing	6 Version releases, 48-case test suite, semantic commits	test_suite_audit.py, Git branches
DBMS	3NF Normalization, ACID transactions, WAL mode	5 Relational tables, transaction commit/rollback, WAL pragma	schema_dbms.sql, main.py DB block
Computer Networks	RESTful architecture, HTTP status codes, CORS	Async REST routes, HTTP 200/401/422, CORS middleware	main.py FastAPI endpoints
Operating Systems	Async event loops, non-blocking I/O, singleton caching	Uvicorn ASGI workers, OCR singleton model cache	main.py startup events, asyncio
Data Structures	TF-IDF sparse matrices, DFA regex, queueing models	CSR matrices in MNB, O(N) PII regex, Erlang-C formulas	main.py queue & PII masking routines
Cloud Computing	Decoupled tiers, edge assets, dynamic base URLs	React/Vite SPA bundle, dynamic QR base URL synthesis	frontend/dist, main.py QR engine

Chapter 7: Implementation Details

7.1 Frontend and Backend Architecture

The frontend application is implemented with React 18 and Vite. LanguageContext.jsx exports a useLanguage hook maintaining verified 222-key trilingual parity across English, Hindi, and Marathi. SubmitComplaint.jsx orchestrates browser-level webkitSpeechRecognition listeners with an isolated baseTextRef to prevent repeated speech loops.

The backend is organized in Python_codes/main.py using FastAPI, Uvicorn, and Pydantic schemas. It mounts static evidence file directories, enforces CORS policies, sanitizes PII tokens at ingestion, and streams dynamic ReportLab PDF receipts.

7.2 Version Control and Git Evolution

- **Version_0:** Baseline multilingual setup, initial SQLite schema, and Hinglish validation.
- **Version_1:** 15,000 grievance dataset integration, NLP categorization, and animated Nova assistant.
- **Version_2:** Full Marathi localization expansion and frontend build stabilization.
- **Version_3:** Explainable AI (XAI) reasoning traces, customer resolution memos, and queue filtering.
- **Version_4:** Bank performance leaderboard and navigation streamlining.
- **Version_5:** Complete 222/222 key trilingual parity, case-insensitive DPDP PAN masking, dynamic QR base URL.
- **Version_6:** Master Audit Certified: zero-latency voice dictation, clean production Vite build, 48/48 test pass.

Chapter 8: Scalability, Performance and Resource Management

8.1 Endpoint Latency Profile

Endpoint	Min Latency	Median (p50)	p99 Latency	Operational Status
GET /api/health	1.8 ms	3.2 ms	8.4 ms	Optimal
POST /api/complaints/submit	18.5 ms	24.1 ms	48.6 ms	Optimal
GET /api/complaints/{id}	3.4 ms	5.1 ms	12.0 ms	Optimal
GET /api/complaints/{id}/pdf	45.2 ms	62.8 ms	110.4 ms	Optimal
GET /api/analytics/queue-status	12.0 ms	28.4 ms	64.2 ms	Cached (96.7% faster)
POST /api/verify-evidence	340.0 ms	480.0 ms	820.0 ms	Compute-Bound

8.2 Bottleneck Resolutions

- **Analytics Optimization:** Full table scans over 15k records initially caused 855.4 ms latency. Introducing composite database indexing and a 5-second TTL memory cache cut latency to 28.4 ms (96.7% speedup).
- **Bundle Splitting:** Vite code-splitting with dynamic React.lazy imports reduced initial bundle size to 348 KB, achieving sub-second Time-to-Interactive (TTI = 0.85s).

Chapter 9: Testing and Validation

9.1 Master Automated Verification Matrix

Test ID	Req ID	Scenario	Input	Expected Output	Status
TC-01	FR-01	Backend Health Probe	GET /api/health	HTTP 200, status ok	**PASS**
TC-02	FR-01	Complaint Submission	Valid complaint JSON	HTTP 200, generated ID	**PASS**
TC-03	FR-02	NLP Category Inference	ATM dispensed no cash	Category ATM Withdrawal	**PASS**
TC-04	FR-04	16-Digit Card Masking	Card 4111 2222 3333 4444	Masked: XXXX-XXXX-XXXX-XXXX	**PASS**
TC-05	FR-04	12-Digit Aadhaar Masking	Aadhaar 1234 5678 9012	Masked: XXXX-XXXX-XXXX	**PASS**
TC-06	FR-04	Uppercase PAN Masking	PAN ABCDE1234F	Masked: XXXXX0000X	**PASS**
TC-07	FR-04	Lowercase PAN Masking	PAN abcde1234f	Masked: XXXXX0000X	**PASS**
TC-08	FR-05	Statutory PDF Download	GET /complaints/{id}/pdf	Valid PDF stream	**PASS**
TC-09	FR-05	Dynamic QR URL Resolution	Request with Host header	Live URL encoded	**PASS**
TC-10	FR-06	OCR Amount Entity Match	Receipt matching Rs 5000	Amount badge flagged	**PASS**
TC-11	FR-06	OCR Date Entity Match	Receipt matching txn date	Date badge flagged	**PASS**
TC-12	FR-06	OCR Bank Name Match	Receipt matching bank name	Bank badge flagged	**PASS**

TC-13	FR-06	Credibility Scoring	All 4 entities matched	Score = 100%	**PASS**
TC-14	FR-07	M/M/c Dwell Time Math	Arrival=15, Service=5, c=4	Valid Wq calculated	**PASS**
TC-15	FR-07	Queue Overload Handling	Arrival=25, Service=5, c=4	Graceful limit handling	**PASS**
TC-16	FR-08	SLA Countdown Trigger	Pending complaint > 72h	High SLA risk alert	**PASS**
TC-17	FR-09	Resolution Memo Dispatch	Officer signature & memo	Status: Resolved	**PASS**
TC-18	FR-10	Trilingual Parity: English	Scan 222 language keys	222/222 keys valid	**PASS**
TC-19	FR-10	Trilingual Parity: Hindi	Scan 222 language keys	222/222 keys valid	**PASS**
TC-20	FR-10	Trilingual Parity: Marathi	Scan 222 language keys	222/222 keys valid	**PASS**
TC-21	FR-03	Voice Non-Duplication	Continuous speech stream	Zero duplicated text	**PASS**
TC-22	FR-01	Empty Payload Rejection	Empty JSON {}	HTTP 422 rejected	**PASS**
TC-23	FR-01	Negative Amount Handling	Amount = -500.00	HTTP 422 rejected	**PASS**
TC-24	FR-01	Non-Existent Complaint ID	GET /complaints/FAKE-ID	HTTP 404 returned	**PASS**
TC-25	FR-09	Unauthorized Escalation	Customer calls Officer API	HTTP 401/403 blocked	**PASS**
TC-26-48	Misc	Stress, UI & Boundary Checks	23 boundary/stress tests	All passed specifications	**PASS**

Chapter 10: Results, Demonstration and Discussion

10.1 Functional Demonstration

When a Marathi-speaking user dictates: 'Maza SBI account madhun 5000 rupaye debit zale pan ATM madhun paise nighale nahit. Maza card 4111 2222 3333 4444 ahe', the system streams speech without lag, redacts the card to 'XXXX-XXXX-XXXX-XXXX', classifies the dispute as 'ATM Withdrawal Failure' (94.6% confidence), generates statutory receipt GIQ-2026-X with a dynamic QR code, and registers the complaint in the ATM queue.

10.2 Achievement of Objectives

Objective	Implementation Module	Validation Evidence
Autonomous NLP Triage	Scikit-Learn MNB on 15k dataset	89.2% accuracy; TC-03 pass
Continuous Voice Dictation	W3C Web Speech isolated-buffer architecture	Zero looping; TC-21 pass
DPDP Act 2023 Masking	Case-insensitive Card/Aadhaar/PAN regex	100% scrubbed; TC-04 to TC-07
Statutory PDF + QR Receipt	ReportLab dynamic URL QR generator	Valid PDF; TC-08, TC-09 pass
Predictive Queue Analytics	Erlang-C closed-form M/M/c math engine	Real-time Wq & utilization; TC-14, TC-15
Explainable Evidence OCR	EasyOCR + 4-point entity matcher	Credibility score; TC-10 to TC-13

Chapter 11: Conclusion and Future Scope

11.1 Conclusion

GrievanceIQ delivers an autonomous, trilingual, DPDP-compliant banking grievance redressal platform. By coupling lightweight machine learning (MNB), continuous voice dictation, verifiable QR receipts, and M/M/c queueing theory, the platform solves major real-world banking bottlenecks. All 48 test cases passed cleanly, and the production codebase is live on GitHub under Version_6 releases.

11.2 Future Enhancements

- **1. Direct Core Banking Integration:** Connect via ISO 8583 / ISO 20022 interfaces to Infosys Finacle / TCS BaNCS to trigger automated chargeback reversals.
- **2. Conversational Voice Bot:** Evolve the voice dictation engine into a two-way spoken conversational agent to assist non-literate depositors.
- **3. Blockchain Dispute Ledger:** Anchor complaint state transitions into a tamper-evident distributed ledger for statutory banking ombudsman audits.

References

[1] Reserve Bank of India, "Annual Report of the Reserve Bank - Integrated Ombudsman Scheme (RB-IOS) for the Year 2022-2023," RBI Bulletin, Reserve Bank of India, Mumbai, 2023.

[2] Ministry of Law and Justice, Government of India, "The Digital Personal Data Protection Act, 2023 (Act No. 22 of 2023)," The Gazette of India, Extraordinary, Part II, Section 1, August 2023.

[3] F. Pedregosa, G. Varoquaux, A. Gramfort, et al., "Scikit-learn: Machine Learning in Python," Journal of Machine Learning Research, vol. 12, pp. 2825-2830, 2011.

[4] L. Kleinrock, Queueing Systems, Volume 1: Theory. New York: John Wiley & Sons, 1975.

[5] G. Shires and H. Wennborg, "Web Speech API Specification," W3C Community Group Final Report, World Wide Web Consortium (W3C), 2012.

[6] International Organization for Standardization, "ISO/IEC 18004:2015 Information Technology - Automatic Identification and Data Capture Techniques - QR Code Bar Code Symbology Specification," Geneva, Switzerland, 2015.

[7] A. McCallum and K. Nigam, "A Comparison of Event Models for Naive Bayes Text Classification," in AAAI-98 Workshop on Learning for Text Categorization, vol. 752, pp. 41-48, 1998.

[8] R. S. Pressman and B. R. Maxim, Software Engineering: A Practitioner's Approach, 8th ed. McGraw-Hill Education, 2014.

[9] A. Silberschatz, H. F. Korth, and S. Sudarshan, Database System Concepts, 7th ed. McGraw-Hill Education, 2019.

[10] A. S. Tanenbaum and H. Bos, Modern Operating Systems, 4th ed. Pearson, 2014.

[11] J. F. Kurose and K. W. Ross, Computer Networking: A Top-Down Approach, 8th ed. Pearson, 2021.

[12] T. H. Cormen, C. E. Leiserson, R. L. Rivest, and C. Stein, Introduction to Algorithms, 3rd ed. MIT Press, 2009.

[13] S. Ramirez, "FastAPI: High Performance Web Framework for Modern Python," 2023. [Online]. Available: https://fastapi.tiangolo.com

[14] Meta Open Source, "React: The Library for Web and Native User Interfaces," 2024. [Online]. Available: https://react.dev

[15] Department of Administrative Reforms and Public Grievances, "Monthly Report on Centralized Public Grievance Redress and Monitoring System (CPGRAMS)," Ministry of Personnel, Public Grievances and Pensions, Government of India, 2022.